

Year 7 • Lesson 10.05 The mean and mode: answers (teacher copy)

10.05 The mean and mode (booklet p. 3)

Warm-up

1. 30
2. 6
3. 7.43
4. 5

WE DO examples

1. Ex 1: $30 \div 5 = 6$
2. Ex 2: 4
3. Ex 3: mean 7.33, mode 9
4. Ex 4: mean 2.33, mode 2 (total $28 \div 12$)
5. Ex 5: $5 \times 8 = 40$; $40 - 30 = 10$
6. Ex 6: mean 17.33, mode 12; the mode (12) is better because 44 is an outlier

Set A

1. 4
2. 7
3. 4
4. 20
5. 5
6. 6
7. 5
8. 15
9. 6
10. 10

Set B

1. 3
2. 4
3. 2
4. 6
5. 10
6. 9
7. no mode
8. 2 and 6
9. red
10. 25

Set C

1. mean 4.71, mode 5
2. mean 4.5, mode 2
3. mean 12.83, mode 15
4. mean 4, mode no mode
5. mean 22, mode 21 and 24
6. mean 2, mode 2
7. mean 20, mode 20
8. mean 2.1, mode 2
9. mean 1.3, mode 1
10. mean 151.67, mode no mode

Set D

1. $24 - 19 = 5$
2. $360 - 293 = 67$
3. e.g. 4, 4, 5, 8, 9
4. mean 12, mode 6; the mode, because 40 is an outlier
5. Mean goes from 7 to 9.17 (up about 2.17)
6. $5 \times 8 = 40$; $40 - 30 = 10$
7. $90 - 70 = 20$
8. The mode (size 8), because it is the size sold most often
9. 16, 17, 18
10. Both means are 16; A has mode 15, B has no mode

Extension

1. $650 + 15 = 665$; $665 \div 10 = 66.5$
2. $96 - 14 = 82$; $82 \div 6 = 13.67$
3. Yes, e.g. 1, 2, 3, 4 has mean 2.5

Exit ticket

1. Level 1: 8
2. Level 2: mean 5.5, mode 7
3. Level 3: $36 - 24 = 12$